Nexify Hackathon '26

Nexify'26

- **Project Name -** AI-powered Smart Agriculture Advisor: boosting Crop Yields and Farmer Decision-Making
- **Team Name:** AI (Aloo Intelligence)
- **Team Leader:** Nitish Sheoran
- **Members Name:** Narayan Prasad
  Sonal
- **Organisation Name -** Gautam Buddha University

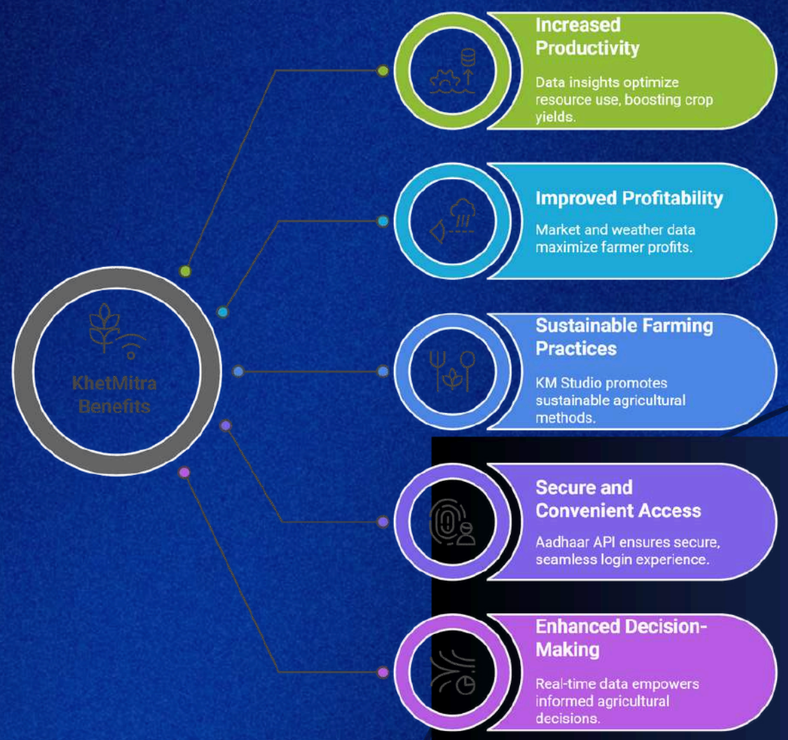
KHETMITRA : AI-Powered Personal Farming Assistant for Farmers

Your AI-powered farming companion for smarter, sustainable harvests

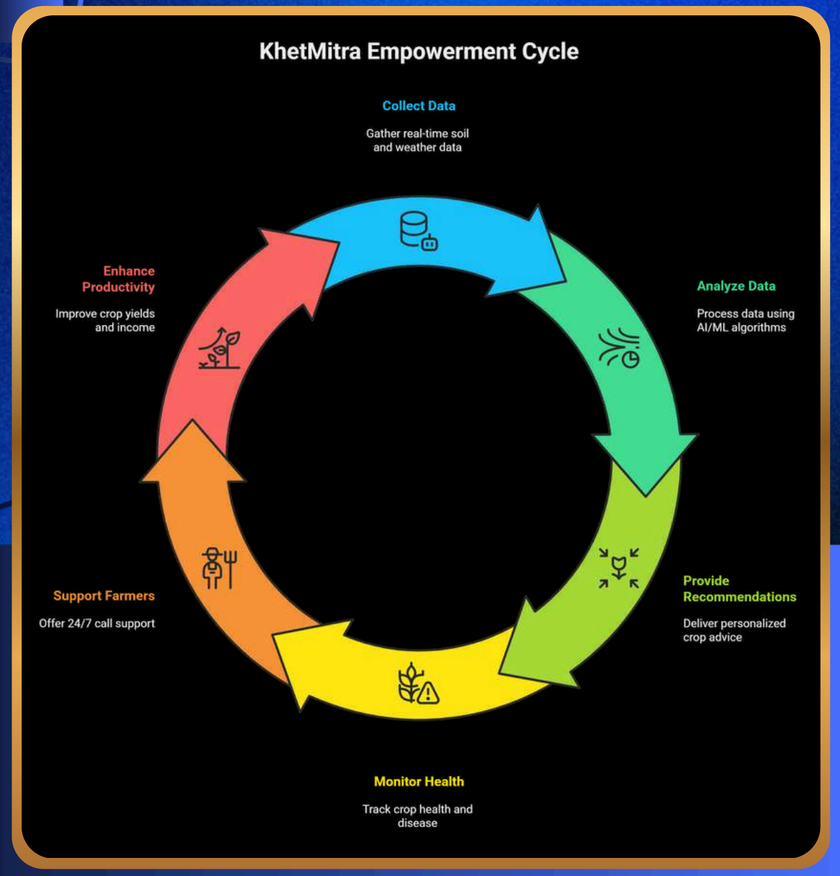
DETAILED EXPLANATION OF THE PROPOSED SOLUTION

KhetMitra uses soil sensors with an **ESP32 and SIM module** to collect real-time soil data, combined with weather, satellite data, soil maps, crop history, and current mandi bhav. AI/ML analyzes this data to recommend crops, predict yield and profit, monitor crop health, detect diseases, and support sustainable farming. The platform offers an **"end-to-end solution—from crop selection to selling"**—helping farmers make smarter decisions and boost productivity. It supports multiple languages, works offline for areas with poor connectivity, and provides 24/7 call support to assist farmers anytime.

INNOVATION & UNIQUENESS OF THE SOLUTION



HOW IT ADDRESSES THE PROBLEM



TECHNICAL APPROACH

TECHNOLOGIES TO BE USED

SOFTWARE:

python TensorFlow Keras
AI/ML

SOILGRIDS Bhuvan Indian Geo-Platform of ISRO OpenWeather
Data Sources & APIs

Kotlin
Android

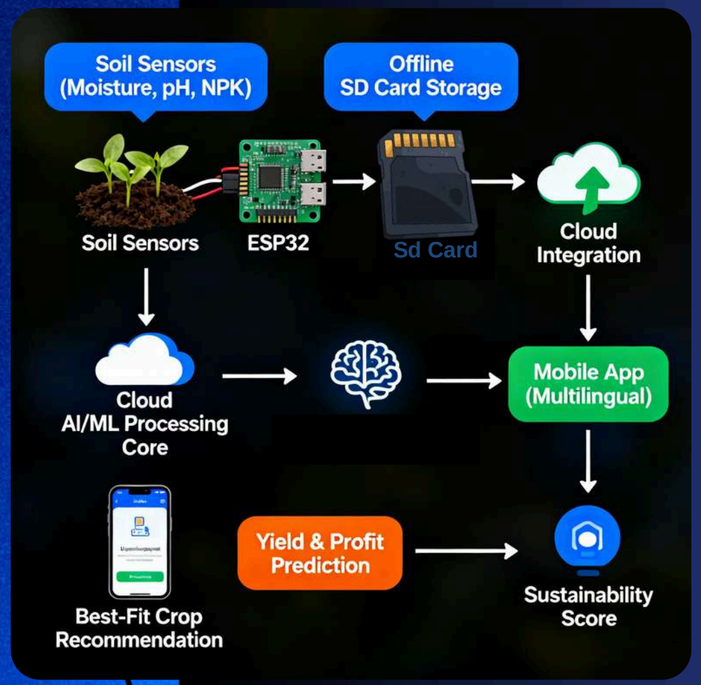
REST API mongoDB Atlas node Express JS

React JS Material UI Redux
Web dashboard UI

HARDWARE:

- Soil Sensors (pH, Moisture, Nutrients)
- ESP32 Microcontroller
- SD Card Module
- IoT connectivity (Wi-Fi&4G).

METHODOLOGY & PROCESS OF IMPLEMENTATION



Website:-https://www.khetmitra.live/

FEASIBILITY & VIABILITY

FEASIBILITY ANALYSIS

Cost-effective at ₹4,500
Government policy alignment (PM-KISAN)
Multiple revenue streams & scalable model

CHALLENGES & RISKS

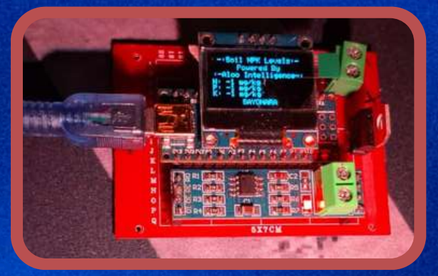
CHALLENGES & RISKS

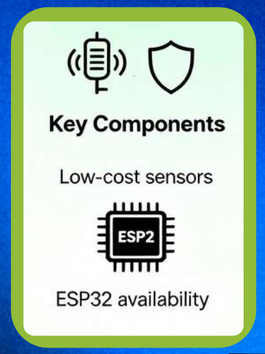
- Calibration accuracy
- Rural connectivity gaps
- Data privacy issues

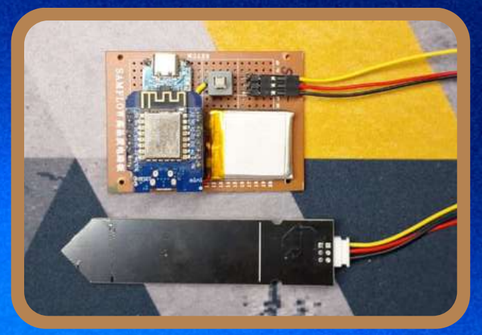
STRATEGIES

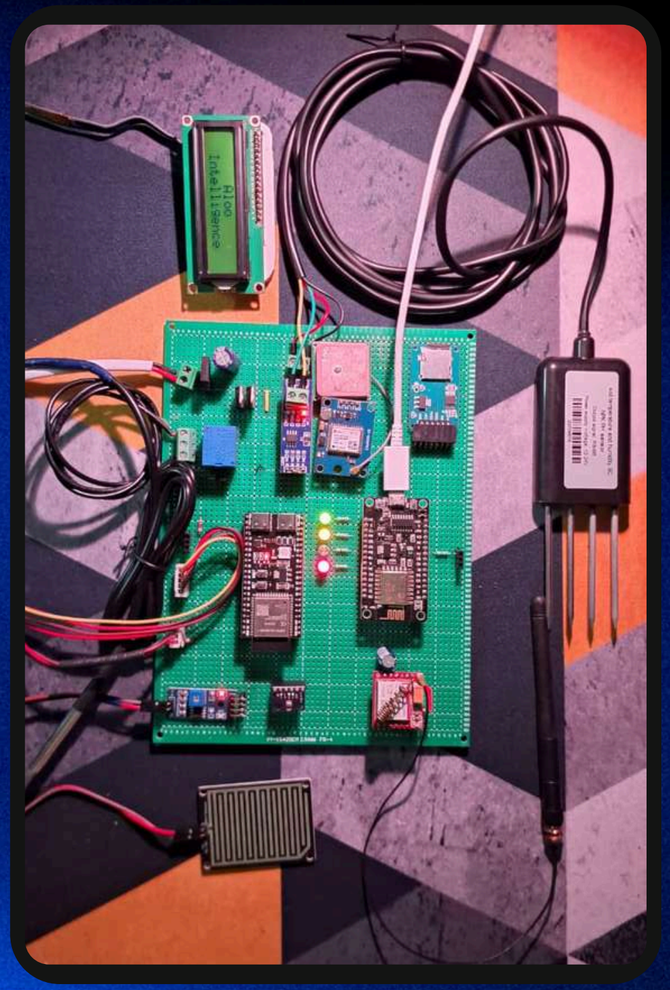
- Field validation + calibration
- Offline data logging (ESP32 + SD card)
- Secure APIs + cloud encryption

PROTOTYPE IMAGES









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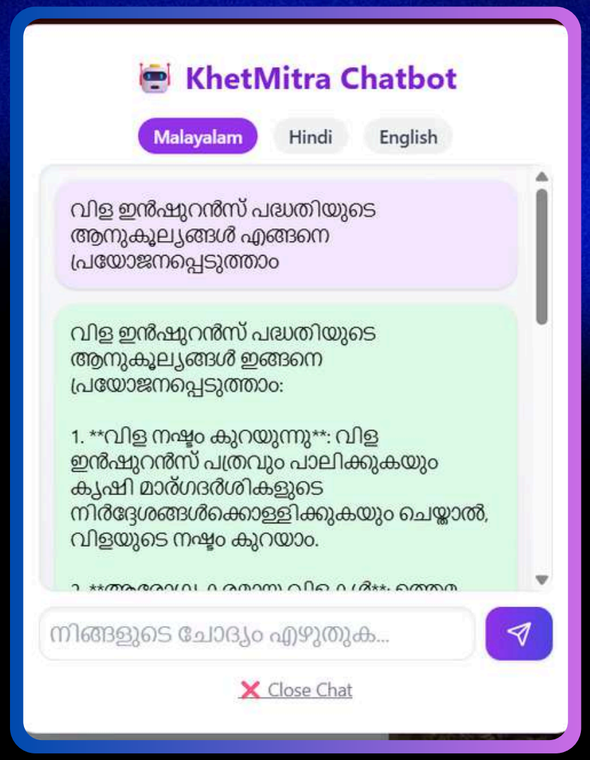
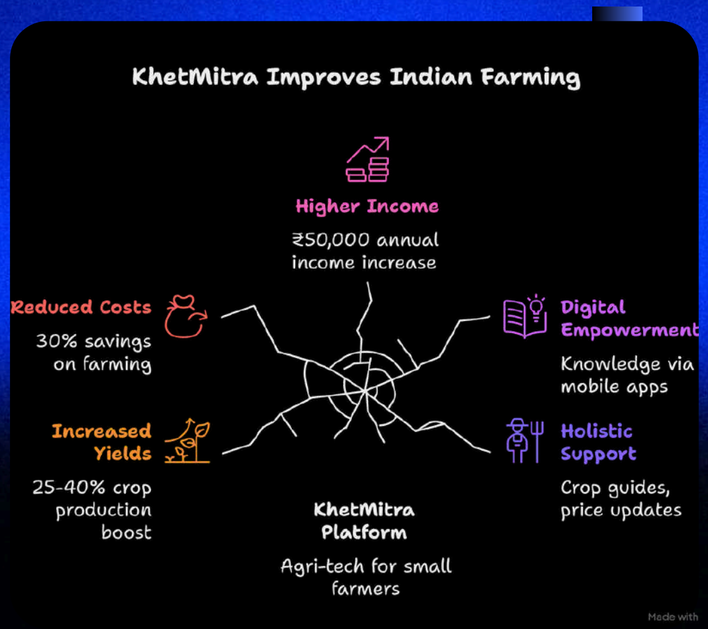
IMPACT AND BENEFITS

BENIFIT OF THE SOLUTION



POTENTIAL IMPACT

- Data-driven crop recommendations & yield prediction
- AI-backed insights replacing guesswork
- Smarter irrigation & fertilizer planning
- Real-time alerts (weather, pests, soil health, Grazing & High wind)
- Market linkage support for better income



IMPORTANT DETAILS

- Real-time IoT-driven farming decisions using live NPK, soil moisture, and environmental sensor data
- AI-powered crop & fertilizer recommendations based on actual soil health, not generic datasets
- Context-aware intelligence (location, season, crop stage) for precise and personalized guidance
- End-to-end system: IoT hardware → backend APIs → AI decision engine → farmer-friendly interface
- Scalable & low-cost solution designed especially for small and marginal farmers
- Actionable insights, not just data — clear “what to grow, when to irrigate, and how much fertilizer”

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THANK YOU